Urban Heat Prediction And Mapping

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***Abstract*—The increasing rate of urbanization and changes in land use pose significant environmental challenges, including the Urban Heat Island (UHI) effect, which causes an increase in land surface temperature (LST) in affected urban areas due to built-up area expansion, reduced vegetation cover, and increased human activities.**

As such, accurate prediction and spatial mapping of urban heat are critical to implement effective urban design and mitigation.

This paper gives an integrated method of urban heat prediction and the development of urban heat maps that utilize remote sensing data, geospatial analysis and machine learning techniques. The main parameters that were investigated include land use/land cover, vegetation index, built-up index, and climate parameters and their impact on temperature variation, machine learning models were used to determine land surface temperature, and geographic information system (GIS) tools were applied in generating spatial heat maps in order to determine areas of urban heat concentration.This study shows that it is the possibility to much more precisely predict, to obtain a clear view of the distributions of heat in urban areas, that the influence of urban morphologies/environmental features on the patterns of temperature, and that it is this data driven decision making that leads to sustainable urban development and effective heat mitigation planning that is achieved in full by this study.

***Index Terms*—Urban Heat Island , Land Surface Temperature , Remote Sensing, Geographic Information System , Machine Learning, Urban Heat Prediction, Heat Mapping, Land Use/Land Cover**

I. Introduction

The high rate of urbanization has greatly changed the natural landscapes into built-up environments, which have resulted in a number of environmental challenges. Among the most serious problems, the rise in temperature in cities in comparison to the surrounding rural areas can be mentioned, which is known as the Urban Heat Island (UHI) effect. The factors that primarily lead to this phenomenon include dense infrastructure, less vegetation cover, more impervious surfaces and anthropogenic heat emissions.

Urban heat has serious impacts on human health, energy consumption, and environmental sustainability. Studies have shown that land use and land cover (LULC) changes play a major role in increasing land surface temperature (LST) in cities. As an example, seasonal changes and vegetation cover are strong factors that affect the thermal conditions in urban areas, particularly developing ones like India [1]. On the same note, extensive urban growth has been cited as a primary cause of the increased urban temperatures and the increased effects of UHI [2].

As such, traditional surface temperature measurement techniques using weather stations are limited in their ability to capture the spatial heterogeneity present across urban spaces. Hence, remote sensing and geospatial methods are extensively used to assess and map the land surface temperature. Remote sensing from satellites offers an effective and precise means of investigating temperature distribution to determine urban heat island hot spots [3].

Machine learning methods have gained momentum in the last decades. Data-driven approaches like regression and artificial neural networks (ANN) have been found to be effective in estimating land surface temperature with high accuracy [4]. Other AI-based advanced machine learning approaches such as random forest have also been applied to urban heat severity analysis and the identification of influencing factors [5], [6].

Brilliantly, there have been some research works to understand the factors that enforce urban heat. Among others urban morphology, spatial structure and built-up density have been named important factors driving intra-urban temperature heterogeneity [7], [8]. Additionally, longitudinal studies have shown that urbanization produces temperature fluctuations with substantial increases across multi-year time frames, and that predictive modeling and sustainable management of urbanization would be required to mitigate these impacts. [9] Researchers have begun to gain some important insights into urban heat; however, there is still ample room for further discovery. In addition to a growing body of work focused on predicting urban heat and/or measuring it, there has not yet been a focus on integrating these two approaches together. One potential approach to tackling this problem is to utilize

both machine learning (to help predict urban heat) and spatial mapping (to illustrate where the heat is occurring). Integrating these two approaches could result in an improved understanding of how urban heat is spatially organized. To accomplish this objective, this study seeks to establish a comprehensive framework for estimating and mapping urban heat, based on combined geospatial data source(s) and advanced analytical methodologies. Using this method will produce a clearer picture of urban heat, as well as how urban heat is affecting our communities.

II. Related work

Remote sensing, geospatial, and machine learning methods have been employed to pervasively study urban heat and land surface temperature (LST). The primary emphasis of existing studies is on temperature prediction, analysis of the influencing factors, and spatial evaluation of the effects of urban heat island (UHI).

A number of scholars have examined machine learning methods to predict urban heat. Gupta and Aithal [4] constructed a data-driven model with multiple linear regression and artificial neural networks (ANN) to predict land surface temperature, and they found better accuracy of ANN models. In a similar fashion, Mansouri and Erfani [5]used machine learning algorithms to forecast the severity of urban heat island, which demonstrated the usefulness of data-driven methods in determining high-risk areas. Land use and land cover (LULC) alterations are considered to be one of the significant sources of urban heat. Talukdar et al. [1] studied the seasonal changes in LST and demonstrated that vegetation has a significant effect in mitigating urban temperature. Li et al. [2] also pointed out the fact that high urbanization and land conversion exacerbate the UHI and raise the surface temperature.

Spatial analysis of urban heat has been done widely through remote sensing methods and GIS-based methods. The application of satellite images in estimating LST and detecting temperature differences in urban centers was demonstrated by Cetin et al. [3]. Continuous urbanization, as it is observed by long-term projects like Liyaqat et al. [9] causes enduring rises in temperature and predictive analysis is required. Besides prediction, a number of studies have aimed to determine the key drivers of urban heat. He et al. [7]employed the random forest models to determine the effect of urban spatial structure on the temperature distribution, whereas Feng et al. [8] determined the effect of environmental and structural elements on the LST. Another study by Huang et al. [10] also demonstrated that urban morphology is a major factor that influences patterns in UHI.

The latest developments in artificial intelligence have contributed to the improvement of urban heat studies. Merabtine and Snaiki [11] and Ahmed et al. [12] conducted reviews of different machine learning techniques in UHI applications, noting their increasing value. Hoang et al. [6] used geospatial analysis and interpretable machine learning to understand better the correlation between environmental features and changes in temperature.

In spite of these developments, a majority of existing studies concentrate on either prediction or spatial analysis alone. Minimal studies combine both machine learning-based forecasting and GIS-based heat mapping into a single platform. Thus, the proposed study will help to create a combined method of urban heat forecast and spatial mapping to enhance knowledge and promote sustainable urban planning.

III. Methodology

A. Data Acquisition and Preprocessing

Using multi-source datasets from remote sensing platforms, climatic record and geospatial databases, this study geo-analyses. Urban heat analysis requires Land Surface Temperature (LST) data obtained from satellite images, like Landsat, which have thermal infrared channels. It further includes auxiliary datasets such as land use/land cover (LULC) or vegetation indices, built-up indices, and other environmental variables since they greatly affect the urban thermal characteristics [4] [1] [2].

- Filtering out noise and handling missing values
- Remote sensing satellite image geo-corrections
- Feature normalization and scaling
- Joining and resampling of lines for datasets to common resolution

  These activities guarantee that the heterogeneous datasets are homogenous and ready for next sequential processing. Derived indices from remote sensing are calculated during pre-processing to enhance the representation of features [3].

B. Feature Extraction and Model Development

The variables that affect urban heat trends would have been identified through feature extraction techniques. The most important features are as listed below:

- The Normalized Difference Vegetation Index (NDVI), which provides a measure of the vegetation's coverage and thus its cooling capacity.
- The Normalized Difference Built-up Index (NDBI), which represents the degree of urbanization.
- Land surface temperature (LST), which is the dependent variable for prediction.
- Land use and/or Land cover (LULC), which shows the type of land present and classes of urban areas.

  These four have been chosen based on data collected from other published works illustrating how vegetation lowers heat and impervious areas contribute to higher heat [1], [7], [8].

  To establish the relationship between these four features and temperature, multiple machine learning algorithms have used to develop the following relationships:
- Multiple Linear Regression (MLR) is used as the baseline model.
- Random Forest (RF) can capture nonlinearity and importance of features.

- Artificial Neural Networks (ANN) can model complex patterns and provide high prediction accuracy.
  Feature extraction is performed to derive meaningful parameters from the datasets, such as vegetation cover and built-up density, which significantly influence urban heat patterns.
  Multiple models, including regression and Random Forest, are implemented and comparatively analyzed to evaluate their performance in predicting land surface temperature (LST). The comparison is based on standard evaluation metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Coefficient of Determination (R^2). The dataset is divided into training and testing sets to ensure proper model validation and avoid overfitting.
  The trained models are then used to predict temperature values based on user-defined inputs such as location and year. The models were trained using the same datasets that have been developed from the historic datasets and were validated using appropriate means of splitting the datasets. Hyper-parameter optimization was conducted on the models to maximize their performance. Previous studies demonstrated the superiority of Random Forest (RF) and Artificial Neural Networks (ANN) over conventional approaches to urban heat prediction [4], [5] [6].

TABLE I
FEATURES USED FOR URBAN HEAT PREDICTION

S.No	Feature	Description
1	NDVI	Vegetation Index
2	NDBI	Built-up Index
3	LST	Land Surface Temperature
4	LULC	Land Use/Land Cover

C. Prediction, Spatial Mapping, and Evaluation

After training, the temperature models use the trained model to predict the land surface temperatures (LST) for the area being studied. The method for predicting land surface temperature includes combining a variety of environmental/spatial data into the model to produce temperature estimates for specific locations.

The predicted land surface temperatures are then entered into a Geographic Information System (GIS) to create mapped visualizations of the land surface temperature. The mapped visualizations show the area to be classified into three different thermal zones an urban environment based on predicted land surface temperature:

- Hot Zones (i.e., Hot Spots)
- Moderate Zones
- Cool Zones
  Spatial visualizations show the urban heat distribution, allowing urban planners to utilize the maps to assist in making decisions for future mitigation strategies [3], [9].
  Model performance is measured using Statistical Techniques:
- Mean Absolute Error (MAE) — A statistical metric used to evaluate the accuracy of models.
- Root Mean Square error (RMSE)
- Coefficient of Determination (R^2)

These are the three Standard Statistical Methods that will assess model reliability and accuracy of the predictions produced using this model. Once the models are predicted, they will be compared to each other to determine which model has produced the most accurate land surface temperature predictions.

IV. PROPOSED METHOD

This research brings together two very distinct methods of analysis, both statistical regression methods and an ensemble-based Random Forest (RF), to create a systematic, integrated framework for predicting and classifying urban heat islands. The purpose of the landmarking technique introduced in this paper is two-fold; to produce accurate predictions of land surface temperature (LST) and to identify key characteristics of urban heat intensity. The use of the model's ability to analyze the complex interactions between environmental and geo-spatial data and to perform time-series analysis provides an extremely robust and scalable solution for urban heat island analysis.

The architectural model of urban thermal environment prediction system can accept custom inputs (e.g., location, year), allowing areas and years not included in the training set to be predicted dynamically. Besides user inputs, it also includes several environmental parameters, such as land use/land cover (LULC), Normalized Difference Vegetation Index (NDVI) and Normalized Difference Built-up Index (NDBI) as well as historical LSTs. The parameters are examined as they exert a powerful effect on the variance of temperature in urban areas; while vegetation leads to effects of cooling, built-up areas increase heat accumulation. As an initial approach, base relations with respect to temperature are established using regression techniques against the input parameters. Here urban heat is found to be linearly related with physical parameters but in joint context the behaviour pattern of urban heat becomes nonlinear complex Therefore a Random Forest algorithm will be used as main predictive model. Random Forest (RF) is another ensemble learning method, in which multiple decision trees produce output whose results are aggregated to improve prediction accuracy, including variance and overfit.Furthermore, it permits feature importance studies to figure out which factors play a significant role in creating urban heat.

- Purchase of multi-source geospatial and environmental data.
- Prediction of user inputs (location and year) that would be dynamic.
- Data preprocessing, such as noise removal, normalization, spatial alignment.
- Important features like NDVI, NDBI, LULC, and LST were extracted.
- Regression analysis in baseline modelling.

- Predicate Random Forest in advanced prediction and nonlinear modeling.
- Model training and validation with historical data.
- Select an input and predict land surface temperature.
- Heat intensity categories of predicted temperature.
- Data Validity: Depend on explanation to get data validity The estimated temperature values are also categorized into three levels of intensity of heat to make them more easily interpretable. These include:
- High temperature (Hotspots)→ indicated in Red.
- Medium temperature zones → denoted in Orange.
- Low temperature zones→ in Green.

This classification scheme further systematizes the study of urban heat and facilitates action by quickly identifying hot spot areas that need interventions. Accessible heat maps can be generated to provide a clear visualization of spatial distribution of temperature across urbanization by applying GIS-based modeling.

Additionally, the proposed approach enables planners and researchers to make decisions based on forecasts of future scenarios for various years and locations.This integration, which can be predictively modeled, classifies and visualizes within the same framework contributes to usability, thereby making this system a powerhouse in practical work.

To sum up, the current solution has a solid base in terms of its ability to forecast urban heat and map it on the basis of data. This model of combination of regression and Random Forest takes the linear and nonlinear associations, leading to improved prediction and practical information to the sustainable city planning concerning the measures to prevent heat in relation to heat prevention strategies.

V. Model Training and Evaluation

A hybrid prediction framework involving regression techniques and Random Forest algorithm is trained and validated in predicting an accurate estimate of land surface temperature (LST) as a function of multiple environmental and spatial drivers. The NDVI (Normalized Difference Vegetation Index), NDBI (Normalized Difference Built-up Index), LULC (Land use/Land cover) and the past records of LST are on the database used for this study.

Dataset is divided into training and testing subsets (usually 70:30 or 80:20) to validate the learning. We split our data into training and testing sets to train our predictive models and use the test set for performance assessment and generalization capabilities.Regression techniques are first used to seed the relationships between input features and temperature, primarily for linear dependency analysis to mimicking the underlying equations. Even so, given the intricacies and nonlinear relationships of urban heat, Random Forest is still used as a primary predictive model. Random forest is an ensemble learning algorithm that builds several decision trees based on random subsets of the features and data samples and aggregates their output to generate a more accurate and stable prediction.

This method reduces overfitting and enhances robustness, resulting in a good application for various types of environmental data. Additionally, hyperparameter tuning occurs via modifying variables including the number of trees, maximum depth of a tree, and feature selection procedure in order to achieve maximum model performance. There is also an analysis of feature importance in relation to the Random Forest technique for determining which variables affect urban heat changes.Using the trained model, predictions for land surface temperature will be made based on user-defined inputs, such as location and year.

The predicted land surface temperatures will then be compared with observed values to assess the accuracy of the predicted value. Evaluations of model performance are completed on the predictive accuracy of the model using several evaluation metrics. These metrics are: mean absolute error (MAE), which is the average error between actual and predicted land surface temperatures; root mean square error (RMSE), which is the square root of the average squared error between actual land surface temperature and predicted land surface temperature; and coefficient of determination (R^2), which indicates how well the model predicts variance in land surface temperature values. Through these metrics, we will be able to create a complete assessment of accuracy, reliability, and efficiency of the model.The predicted temperature values will also be grouped into one of three heat intensities: high, medium and low. The heat intensity will be presented with a specific colour scheme to represent the locations; red for high temperature heat spots; orange for moderate temperature areas; and green for low temperature areas. The classification of the temperature values provides greater ease of interpretation, as well as the ability to effectively validate how heat has been distributed spatially.There is a found consistency with predicted outcomes and what was actually observed so confirms that the proposed modelling method is robust. The provision of uniform training and careful parameter tuning along with multiple different methods of evaluation that are unambiguous support to produce correct urban heat forecasts, thereby providing for an operationally viable method of developing heat forecasts for cities.

VI. Experimental Results

The Urban Heat Prediction Framework will be assessed in terms of its usefulness through the use of; spatial visualizations, time series forecasting and comparisons across cities.

Fig. 1 displays the prediction summary for the chosen city with background, current and future land surface temperature values. Between 2010, and projected years for the future like 2030 or even as far out as 2038 there is a small increase in temperature leading toward more warning temperatures. Such a shift reflects the ongoing influence of urbanization and environmental factors on intensity of heat in cities. The Random Forest performed well, achieving high prediction performance (high R^2 values and low error metrics), confirming the reliability of this model being able to capture linear and non-linear relationships.

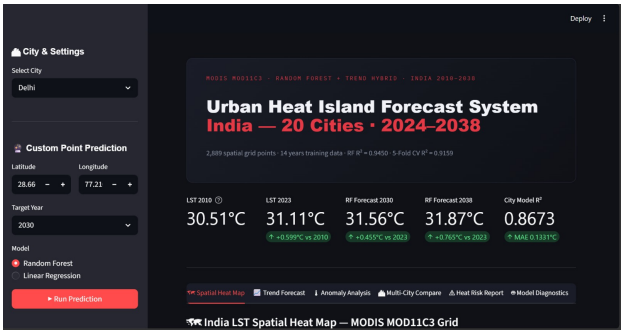

Fig. 1. Prediction summary showing historical and forecasted land surface temperature values.

Fig. 2 The heat map shows the spatial distribution of land surface temperature (LST) in the GIS visualization layer [10]. The figures break this high-temp (red), moderate (orange) and low temp (green) regions. In fact, one can note that densely constructed urban territory is much warmer than green or semiurban areas.

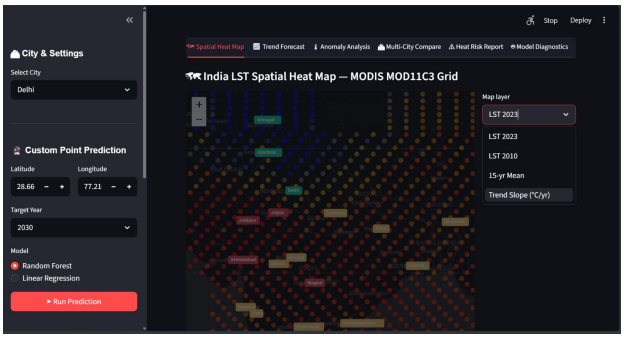

Fig. 2. Spatial heat map showing land surface temperature distribution across the study area.

Fig. 3 presents the historical temporal trend and prediction of land surface temperature. The graph shows a continual trend of rising, showing the increase of temperature in the urban areas. Both historical variations and future projections are well captured by this model which affirms the ability of the model for long-term forecasting.

Fig. 3. Temporal trend and forecast of land surface temperature.

Fig. 4which showcases a comparison for predicted land surface temperature of various cities. The findings indicate substantial differences between urban areas, with certain cities displaying much greater thermal intensity. Such comparisons assist in detecting vulnerable areas and focussing heat mitigation strategies, based on an analysis at the level of city.

Fig. 4. Comparison of land surface temperature across multiple cities.

Overall, the experimental results confirm that the proposed framework provides accurate and reliable predictions of urban heat patterns. The integration of machine learning-based forecasting, spatial heat mapping, and comparative analysis enhances the interpretability of results and supports effective decision-making for sustainable urban planning and heat mitigation strategies.

A comparative analysis of the models indicates that the Random Forest algorithm outperforms the regression model in terms of prediction accuracy and reliability.

TABLE II
COMPARISON OF REGRESSION AND RANDOM FOREST MODELS

Model	MAE	RMSE	R²
Regression	1.25	1.80	0.82
Random Forest	0.85	1.20	0.91

A. Discussion

Results from an urban heat prediction system show that this model can capture spatial and temporal variations of land surface temperature effectively, which is done by combining regression modelling with Random Forest algorithms enabling modelling to complex how environmental parameters like vegetation/built up area have many relationships with temperature.Forecast results show that temperature has gradually increased over time indicating that rapid urbanization and environmental change are responsible for the increase in heat. The system's capability to define input parameters (location/year) means it can be useful for making predictions of urban heat for forecast periods based on different geographic regions.

The heat map shows a clear display of the location of people/temperature distribution in all geographic areas listed. Higher temperature areas of considerable density are mainly found in dense urban environments, whereas areas characterized by significant volumes of vegetation typically have lower temperature overall. These results confirm that vegetation and land use types have a major impact on urban heat formation.In

addition, a trend analysis supports these results by identifying an increase in temperature over time. As such, there is a growing need for heat mitigation strategies and effective urban planning. Additionally, different cities have vastly different temperature levels depending on where they are located and whether they have properly built up their infrastructure and plants.

A key limitation of the model is that the accuracy of predictions is determined by the input data. Therefore, if input data have insufficient technical quality, predictions will also be inaccurate. For example, predicting temperatures during a natural disaster or when the area experiences a severe weather event may not be accurate using this model. Adding additional characteristics to the model, including humidity, population density, and traffic data, could improve the accuracy of the predictions.The system proposed in this report is easy to use, practical, and scalable. It will provide urban planners and policymakers with valuable information and ways to help them better understand and control urban heat, and therefore is a valuable resource for sustainable urban development.

VII. Conclusion

A successful integrated method for urban heat prediction through regression and random forests is presented. The method combines ML, geospatial, and visualization techniques for predicting land surface temperature (LST) and locating urban heat patterns.

The results indicate that the model is capable of accurately capturing spatial and temporal temperatures. The use of multiple environmental factors including vegetation, built-up areas, and land use provides greater accuracy of prediction. By categorizing LST into high, medium, and low using colour coding, this enhances interpretability and makes the model more user-Friendly.Overall, urban temperature data analysis suggests an upward trend over time, indicative of the increasing influence of urbanization. The spatial heat maps, along with the city-to-city analyses, provide information regarding heat distribution in cities and suitable locations for locating urban areas at high risk for heat impacts. This shows the need for data-driven methods to describe and to manage heat in urban environments.

In summary, the proposed algorithms are practical, scalable and suitable for real-world cases. Results like these provide urban planners and policymakers with the data they need to create heat mitigation approaches that also promote sustainable urban development. Future investigations may build on this work by introducing additional features (e.g. humidity, population density, traffic) to enhance prediction and overall generalizability.

Acknowledgment

This research project could not have been realized without the guidance, mentorship and support of multiple faculty members on our journey toward its development, which we are all truly grateful for.

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